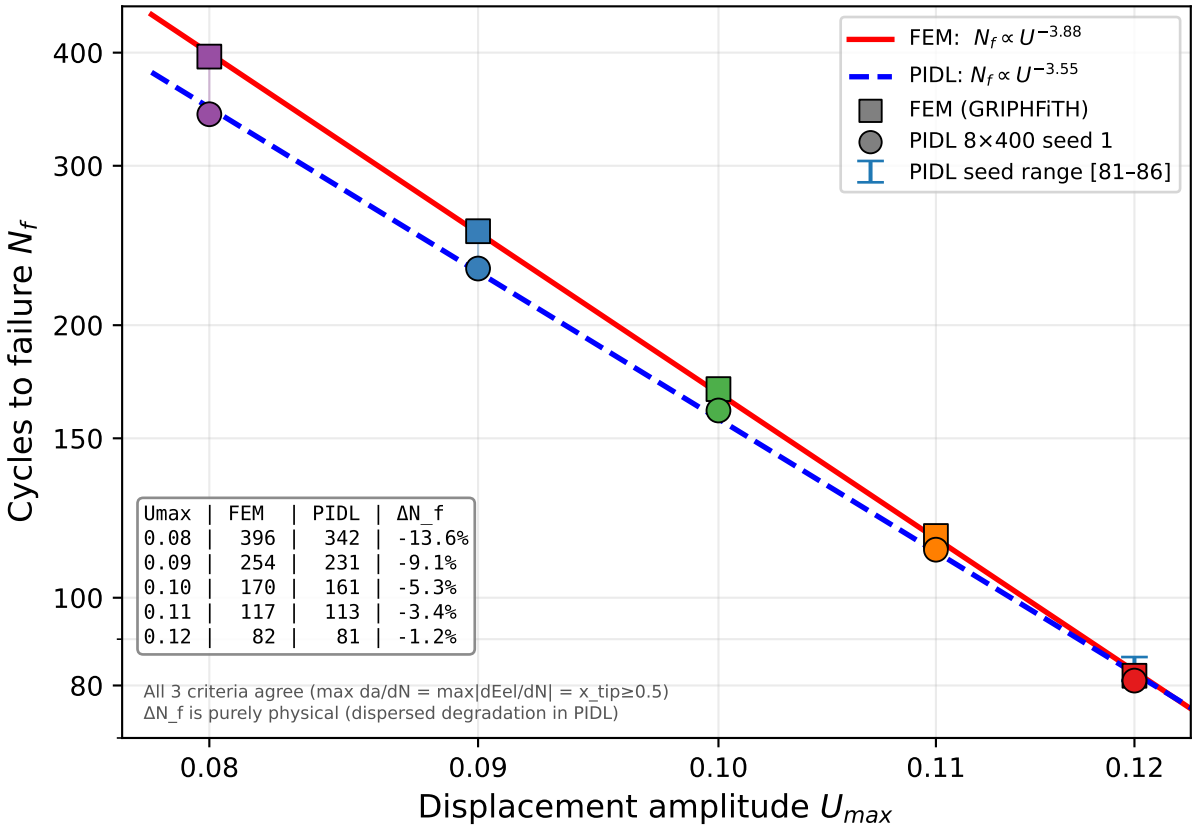
S-N Curve: PIDL 8×400 vs FEM GRIPHFiTH
FEM exponent -3.88 vs PIDL exponent -3.55
FEM: $N_f \propto U^{-3.88}$
PIDL: $N_f \propto U^{-3.55}$
FEM (GRIPHFiTH)
PIDL 8×400 seed 1
PIDL seed range [81–86]
Cycles to failure N_f
400
300
200
150
100
80
0.08
0.09
0.10
0.11
0.12
Displacement amplitude U_{max}
Umax | FEM | PIDL | ΔN_f
0.08 | 396 | 342 | -13.6%
0.09 | 254 | 231 | -9.1%
0.10 | 170 | 161 | -5.3%
0.11 | 117 | 113 | -3.4%
0.12 | 82 | 81 | -1.2%
All 3 criteria agree (max da/dN = max|dEel/dN| = x_tip≥0.5)
ΔN_f is purely physical (dispersed degradation in PIDL)